

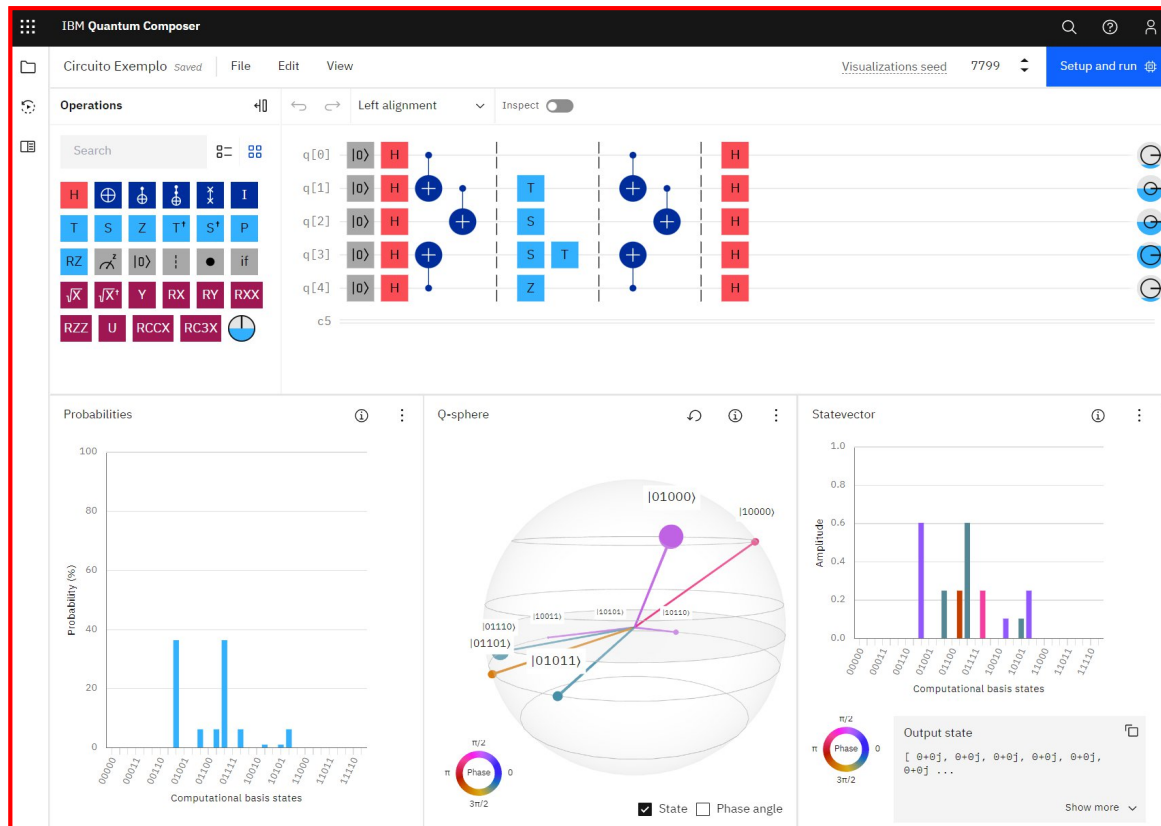
INSTALAÇÃO DO WORKFLOW DE COMPUTAÇÃO QUÂNTICA – UMA ABORDAGEM VIA PLATAFORMA IBM QUANTUM

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Laboratório de Computação Quântica
Instituto de Tecnologia
Universidade Federal do Pará

Plataforma de computação quântica da IBM (*IBM Quantum*)

(1) IBM Quantum Composer

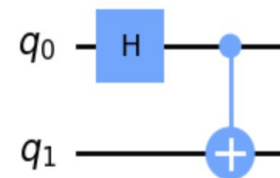


Fonte: IBM Quantum,
<https://quantum.ibm.com>

(2) Quantum Information Science Kit (Qiskit)

```
[1]:  
1  from qiskit import QuantumCircuit  
2  
3  # Create a new circuit with two qubits (first argument) and two classical  
4  # bits (second argument)  
5  qc = QuantumCircuit(2)  
6  
7  # Add a Hadamard gate to qubit 0  
8  qc.h(0)  
9  
10 # Perform a controlled-X gate on qubit 1, controlled by qubit 0  
11 qc.cx(0, 1)  
12  
13 # Return a drawing of the circuit using Matplotlib ("mpl"). This is the  
14 # last line of the cell, so the drawing appears in the cell output.  
15 # Remove the "mpl" argument to get a text drawing.  
16 qc.draw("mpl")
```

Output:



Qiskit is a Python software library for writing, editing, optimizing and running quantum circuits on quantum devices and local simulations.

Fonte: IBM Quantum,
<https://docs.quantum.ibm.com>

Plataforma de computação quântica da IBM (*IBM Quantum*)

<https://quantum.cloud.ibm.com/>



Ciência da informação quântica

Otimização

Simulação hamiltoniana

Aprendizado de máquina

Ciência da informação quântica

Realize pesquisas com o Qiskit no nível de circuitos e portas quânticos, aproveitando as ferramentas de otimização, visualização e execução para transformar ideias fundamentais em experimentos.

Aprendizagem

[Curso de noções básicas de informação quântica](#) →

Explore a pesquisa

Básico



Hello world

Tutorial • Execute seu primeiro circuito quântico e veja os resultados



Biblioteca de circuitos

Documentos • Explore circuitos e componentes quânticos prontos para uso



Entradas e saídas primitivas

Documentos • Saiba como definir entradas e interpretar resultados para primitivos

Plataforma de computação quântica da IBM (*IBM Quantum*)

 IBM Quantum Platform

 Procura

us-east 

[Conectar](#)

IBM Plataforma Quantum

Obtenha acesso aos computadores quânticos IBM, à documentação do Qiskit e aos recursos de aprendizado, tudo em um só lugar.

 Procurar por recursos

Ciência da informação quântica

Otimização

Simulação hamiltoniana

Aprendizado de máquina

Ciência da informação quântica

Realize pesquisas com o Qiskit no nível de circuitos e portas quânticos, aproveitando as ferramentas de otimização, visualização e execução para transformar ideias fundamentais em experimentos.

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Explore a pesquisa

Básico

**Hello world**
Tutorial • Execute seu primeiro circuito quântico e veja os resultados

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Documentos • Saiba como definir entradas e interpretar resultados para primitivos

**Probabilísticos**
Tutorial • Aplique a amplificação de erros probabilísticos para obter melhor precisão

**Documentação**
Procure documentos, APIs e tutoriais

**Funções**
Explore os serviços para acelerar a pesquisa

**Composer**
Construir e executar circuitos graficamente

**Comunicados**
Veja notícias, atualizações e alertas de serviço

Ao final da página, click em Composer.

Plataforma de computação quântica da IBM (*IBM Quantum*)

(1) IBM Quantum Composer

The screenshot shows the IBM Quantum Composer interface. The top navigation bar includes 'IBM Quantum Learning', 'Home', 'Catalog', 'Composer', and 'Lab'. The 'Composer' tab is active, and a 'Menu' button is highlighted with a red '1'. The main workspace is divided into several sections:

- Operations Panel (2):** A grid of quantum gates including H, T, RZ, $\sqrt{X}$, RZZ, $\oplus$, S, $\otimes$, $|0\rangle$, $\sqrt{X}^\dagger$, U, $\otimes$, Z, T † , $\otimes$, Y, RCCX, $\otimes$, $\otimes$, T † , S † , P, $\otimes$, if, RX, RY, RXX, and RC3X. A search bar is at the top of this panel.
- Circuit Editor (3):** A workspace for composing the quantum circuit, showing qubits q[0] through q[4] and a classical bit c2.
- Run Button (6):** A blue button labeled 'Setup and run' with a gear icon.
- Visualizations (4):** Three panels at the bottom: 'Probabilities' (a bar chart showing probability percentages for computational basis states), 'Q-sphere' (a sphere with a point at the top labeled |000000>), and 'Statevector' (a bar chart showing amplitudes for computational basis states). The 'Q-sphere' panel includes a phase angle indicator and a checkbox for 'State'.

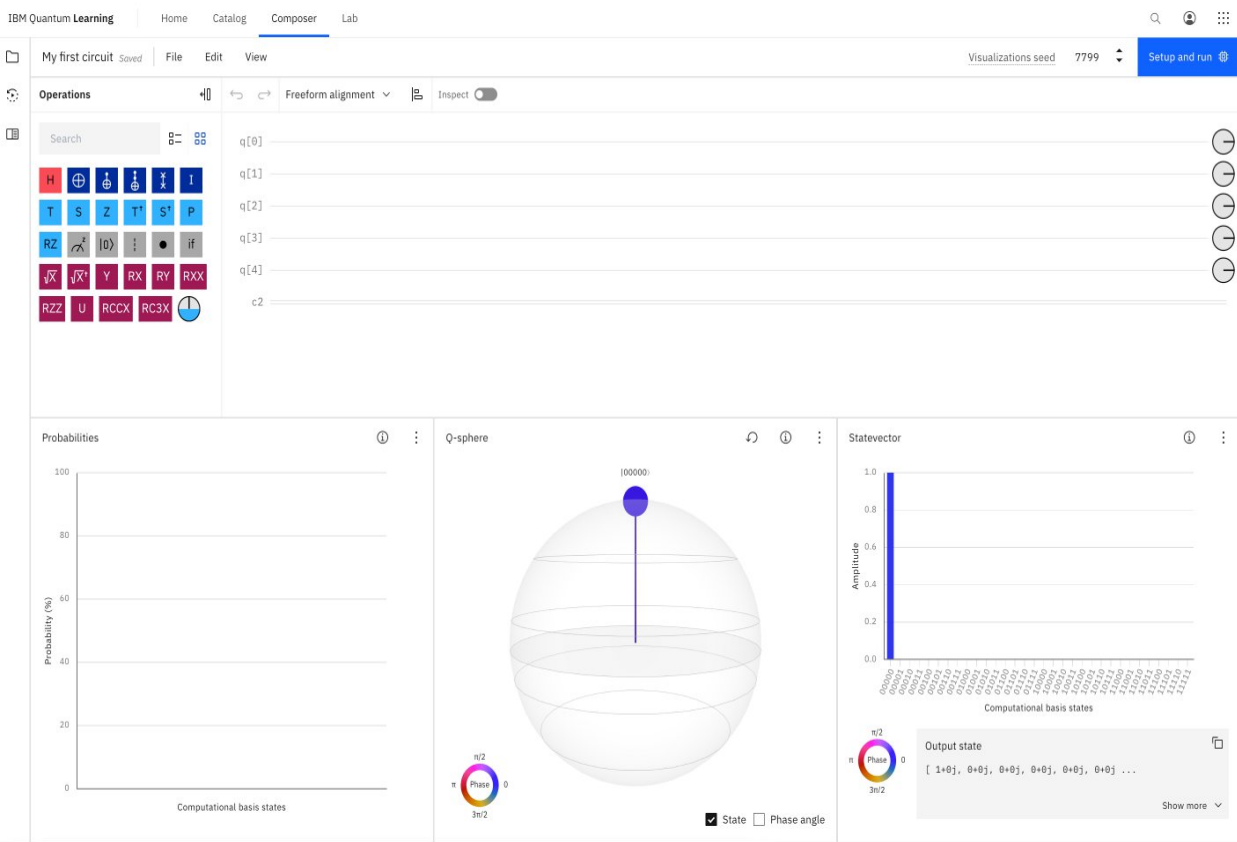
Annotations in the image include:

- Portas lógicas (2):** Logical gates.
- Composição do circuito (3):** Circuit composition.
- Gráficos (5):** Graphics.
- Visualização da Fase dos qubits, superposição e pureza dos Qubits (4):** Visualization of the phase of qubits, superposition, and purity of qubits.
- Run (Devices and Simulators) (6):** Run (Devices and Simulators).

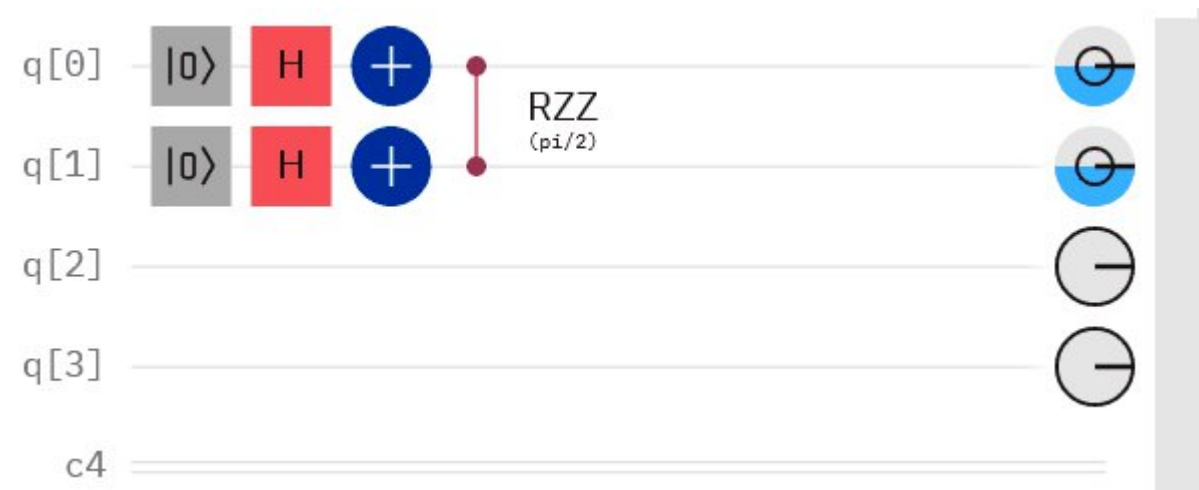
Additional text in the image includes 'Não é Esfera de Bloch!' (Not a Bloch Sphere!) and 'Visualizations seed 7799'.

Plataforma de computação quântica da IBM (*IBM Quantum*)

(1) IBM Quantum Composer



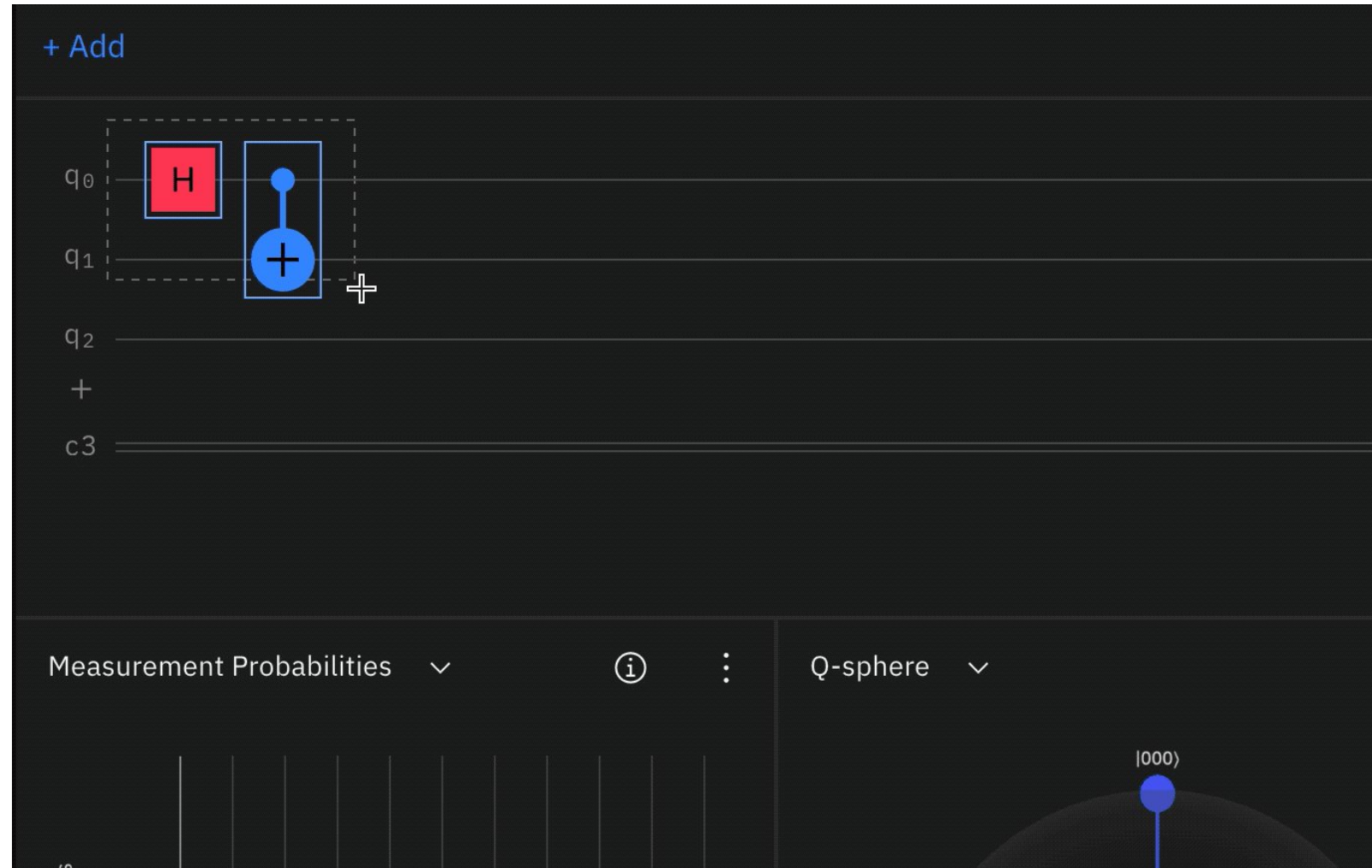
Copie/Cole portas:



Plataforma de computação quântica da IBM (*IBM Quantum*)

(1) IBM Quantum Composer

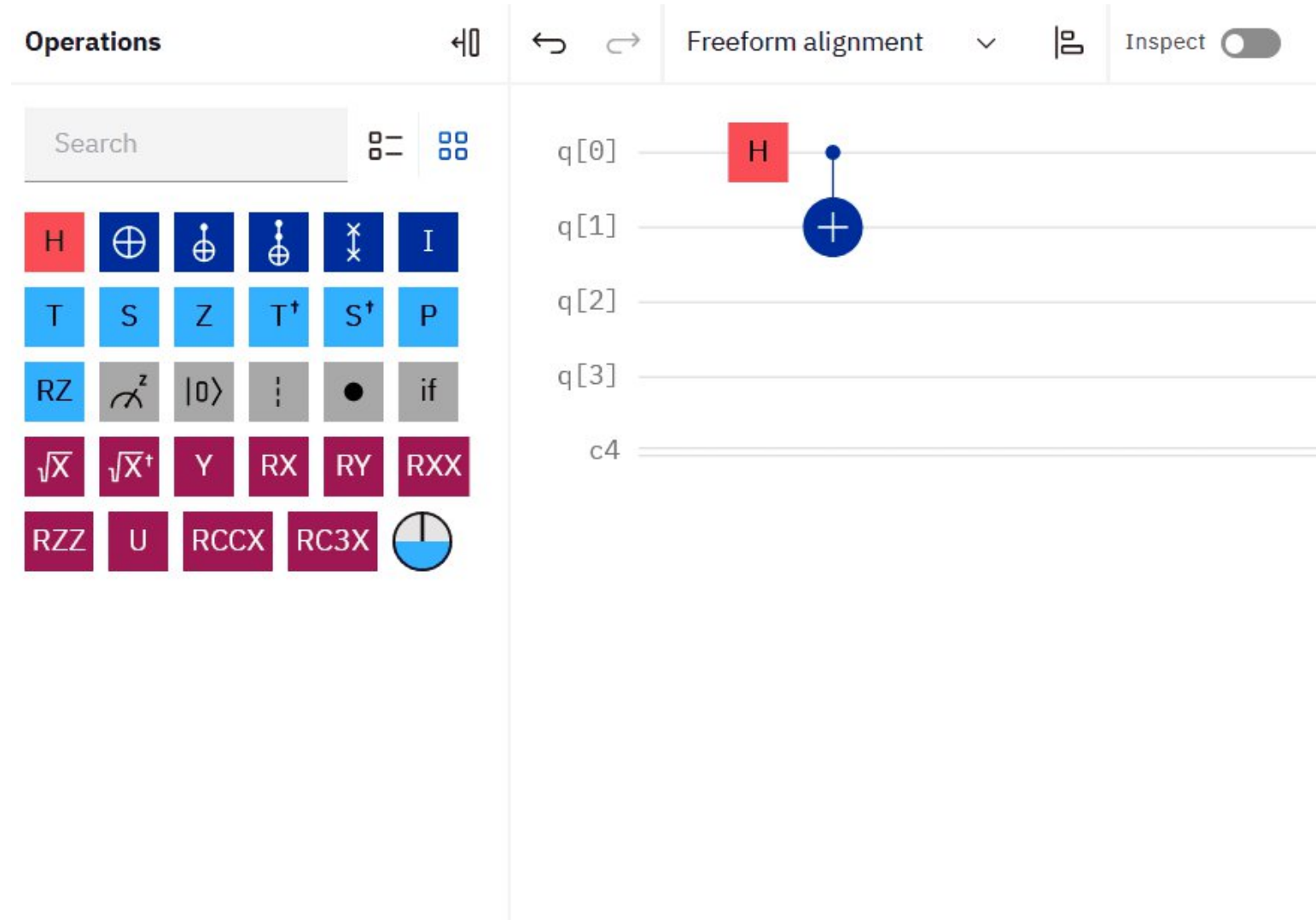
* Para renomear ou excluir uma operação personalizada, clique com o botão direito na operação no catálogo de operações e selecione Renomear ou Excluir.



Plataforma de computação quântica da IBM (*IBM Quantum*)

(1) IBM Quantum Composer

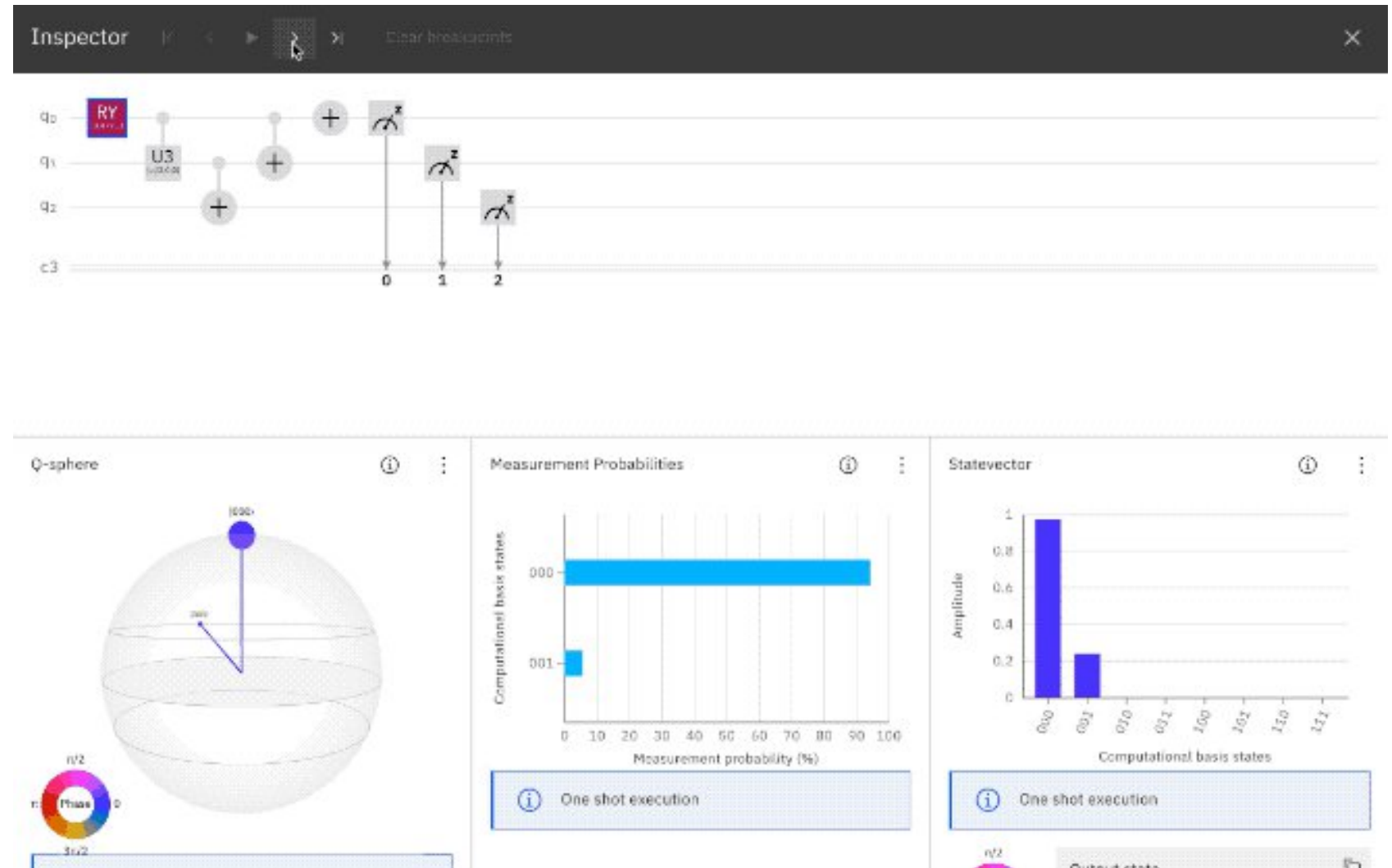
* Arraste o modificador de controle para uma porta para adicionar um controle a ele. Um ponto aparece no qubit de controle e uma linha o conecta ao qubit de destino. Para editar qual qubit é o controle ou destino, clique na porta e selecione o ícone de operação Editar (ou clique duas vezes na porta) para abrir o painel de operação Editar e especifique seus parâmetros.



Plataforma de computação quântica da IBM (*IBM Quantum*)c

(1) IBM Quantum Composer

* Examine seu circuito passo a passo.



2. Protocolo de Codificação Superdensa

Objetivo: A codificação superdensa envolve a transmissão de 2-bits clássicos enviando um único 1-qubit entre duas partes, A (Alice) e B (Bob).

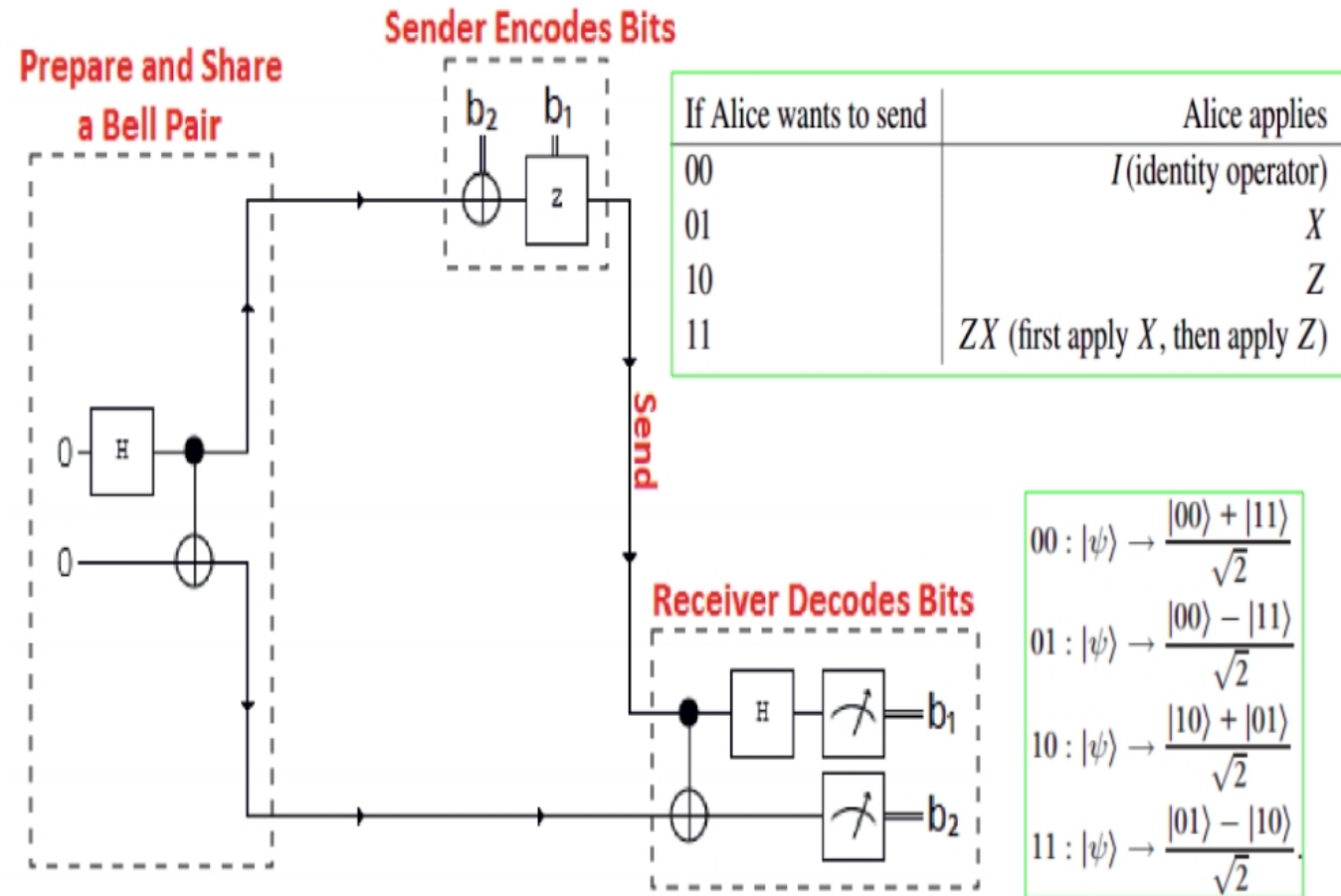
Protocolo da Codificação Superdensa

Passo-1: *Estado Bell bi-partite compartilhado entre as partes A e B.*

Passo-2: Alice realiza operações de 1-qubit (I, X, Z, ZX) para codificar 2-bits, que dependendo do seu interesse, pode ser: 00, 01, 10, 11. OBS.: Alice não interage em nenhum momento com o qubit de Bob.

Passo-3: Alice envia seu qubit para o Bob.

Passo-4: Bob recebe o qubit e decodifica a informação desejada aplicando uma porta Cx(CNOT) e Hadamard(H) sobre os seus qubits e posteriormente realizando medidas na base computacional padrão. Veja a Figura.



2. Codificação Superdensa no Composer (IBM Quantum)

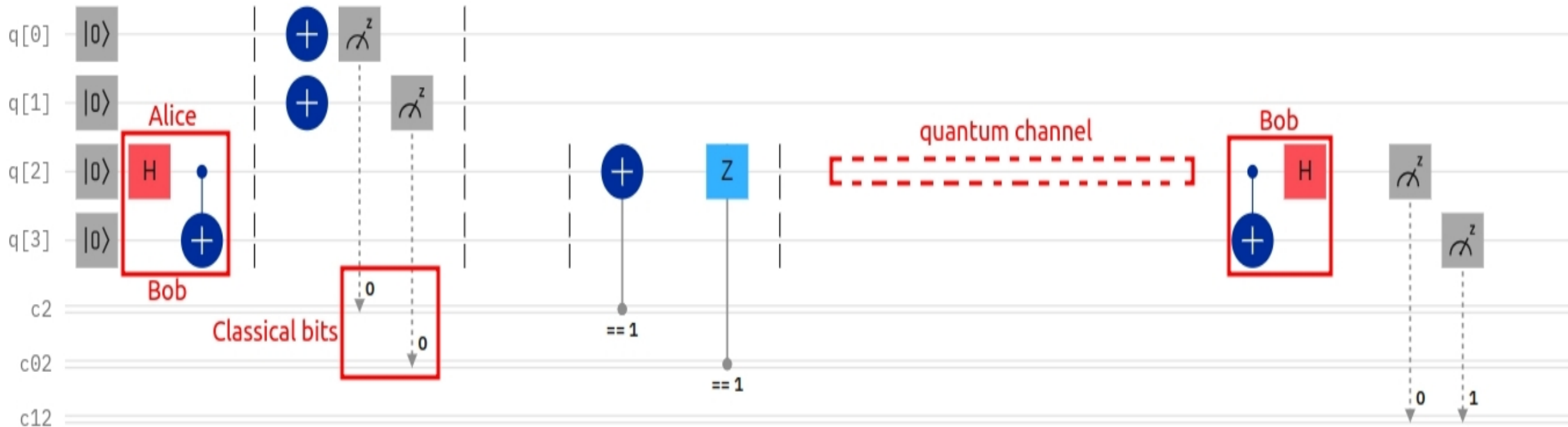
Protocolo da Codificação Superdensa

Passo-1: Estado Bell bi-partite compartilhado entre as partes Alice e Bob. Primeira caixa vermelha da figura abaixo.

Passo-2: Alice realiza operações de 1-qubit (I, X, Z, ZX) para codificar 2-bits, que dependendo do seu interesse, pode ser: 00, 01, 10, 11. OBS.: No caso abaixo, são os bits 11, por isso, 2 NOT's em q0 e q1. Assim, habilitar NOT e Z.

Passo-3: Alice envia seu qubit para o Bob.

Passo-4: Bob recebe o qubit e decodifica a informação desejada aplicando uma porta Cx(CNOT) e Hadamard(H) sobre os seus qubits e posteriormente realizando medidas na base computacional padrão. Veja a Figura.



3. Codificação Superdensa no Composer (IBM Quantum)

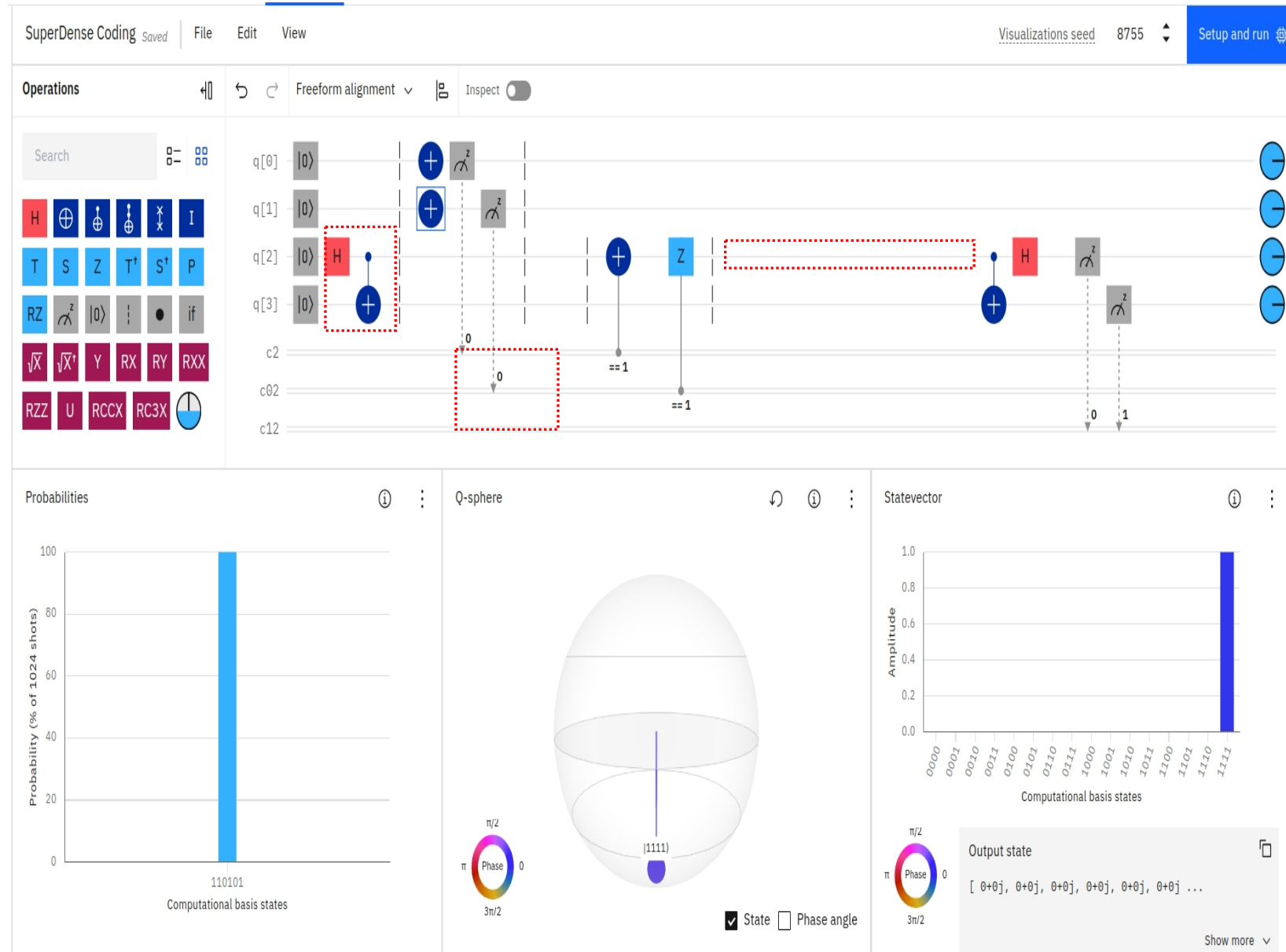
Protocolo da Codificação Superdensa

Passo-1: Estado Bell bi-partite compartilhado entre as partes Alice e Bob. Primeira caixa vermelha da figura abaixo.

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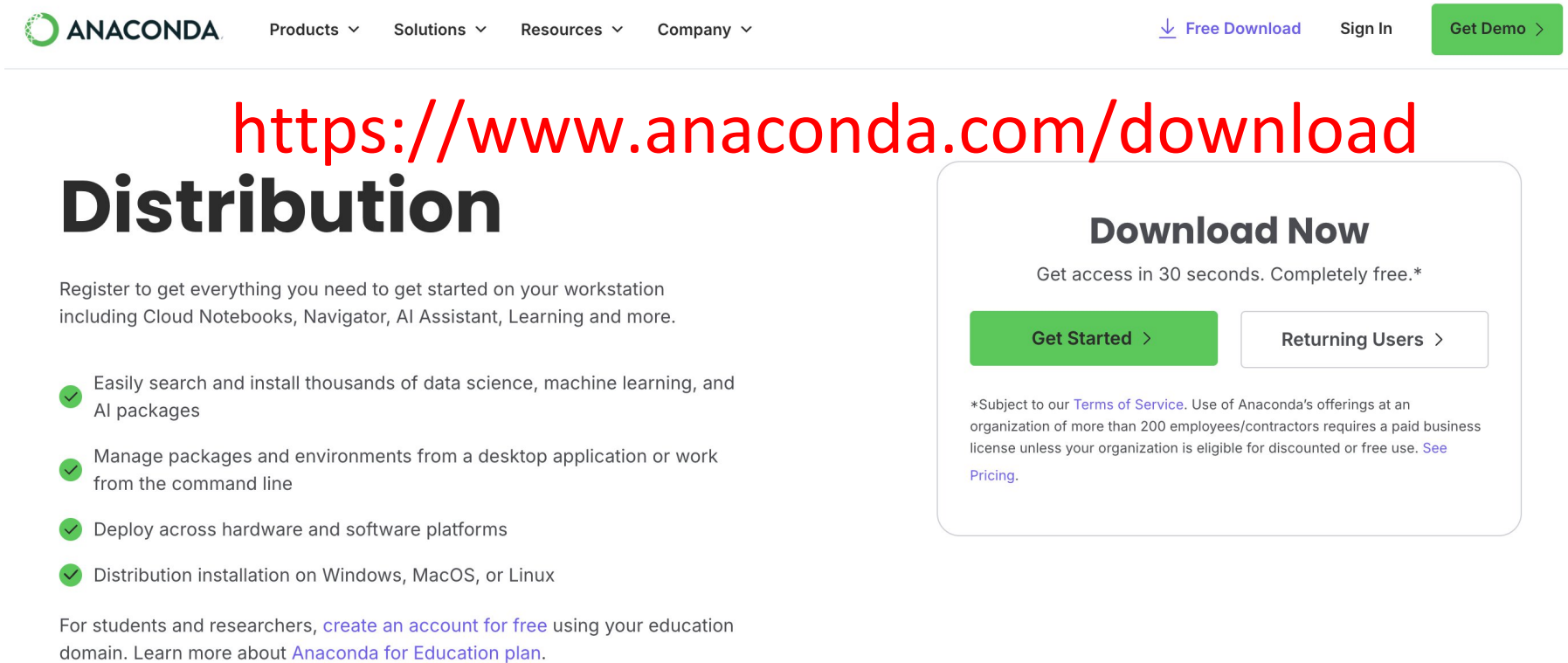
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Plataforma de computação quântica da IBM (*IBM Quantum*)

(2) Framework Qiskit: Instalação do qiskit em um ambiente virtual via conda

1 - Instalando o gerenciador de ambientes Python - Conda via Plataforma Anaconda-Navigator



The screenshot shows the Anaconda website's download page. At the top, there is a navigation bar with the Anaconda logo, links for Products, Solutions, Resources, and Company, and buttons for Free Download, Sign In, and Get Demo. The main heading is "Distribution" in large black font, with the URL "https://www.anaconda.com/download" in red text above it. Below the heading, a paragraph states: "Register to get everything you need to get started on your workstation including Cloud Notebooks, Navigator, AI Assistant, Learning and more." To the right, a "Download Now" section offers access in 30 seconds for free, with buttons for "Get Started" and "Returning Users". A list of four benefits is shown on the left, each preceded by a green checkmark. At the bottom, a note mentions a free account for students and researchers.

ANACONDA Products ▾ Solutions ▾ Resources ▾ Company ▾ [Free Download](#) [Sign In](#) [Get Demo >](#)

https://www.anaconda.com/download

Distribution

Register to get everything you need to get started on your workstation including Cloud Notebooks, Navigator, AI Assistant, Learning and more.

- ✓ Easily search and install thousands of data science, machine learning, and AI packages
- ✓ Manage packages and environments from a desktop application or work from the command line
- ✓ Deploy across hardware and software platforms
- ✓ Distribution installation on Windows, MacOS, or Linux

For students and researchers, [create an account for free](#) using your education domain. Learn more about [Anaconda for Education plan](#).

Download Now

Get access in 30 seconds. Completely free.*

[Get Started >](#) [Returning Users >](#)

*Subject to our [Terms of Service](#). Use of Anaconda's offerings at an organization of more than 200 employees/contractors requires a paid business license unless your organization is eligible for discounted or free use. [See Pricing](#).

(2) Framework Qiskit: Instalação do qiskit em um ambiente virtual via conda

1.1 Instalando o Ambiente Anaconda – Windows, macOS e Linux

Installing on Windows

⚠ Using Anaconda in a commercial setting?

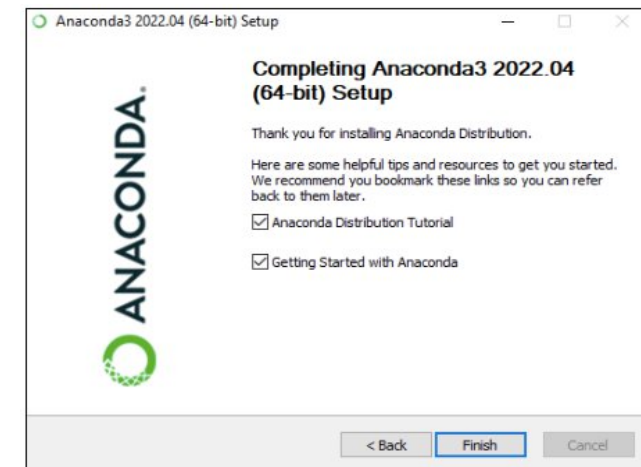
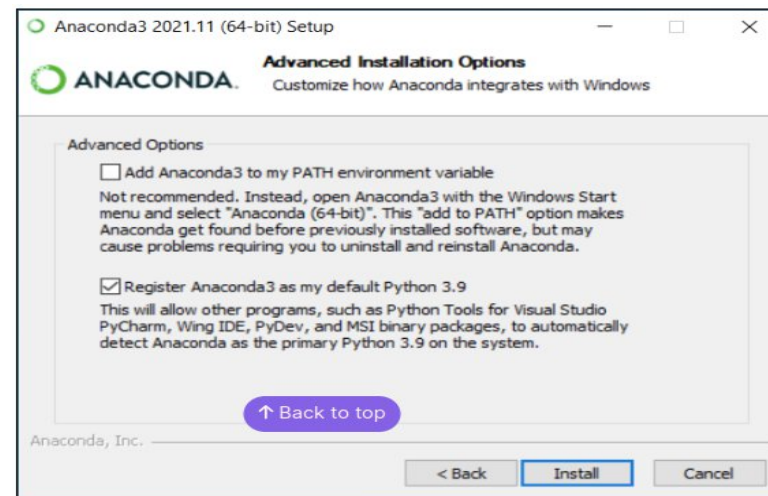
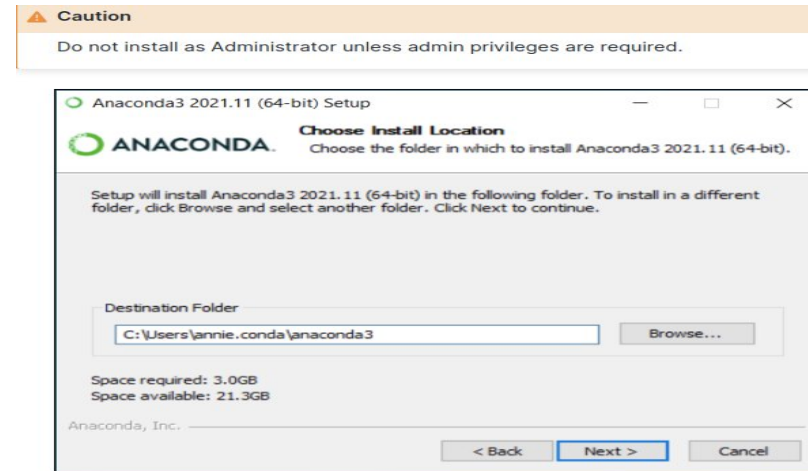
More of a visual learner? Watch the [Installing Anaconda \(Windows\)](#) video in the course linked below!

Anaconda install for Windows

Installation

1. Download the [Anaconda installer](#).
 2. Go to your Downloads folder and double-click the installer to launch. To prevent permission errors, do not launch the installer from the [Favorites folder](#).
- Note**

If you encounter issues during installation, temporarily disable your anti-virus software during install, then re-enable it after the installation concludes. If you installed for all users, uninstall Anaconda and re-install it for your user only.
3. Click **Next**.
 4. Read the licensing terms and click **I Agree**.
 5. It is recommended that you install for **Just Me**, which will install Anaconda Distribution to just the current user account. Only select an install for **All Users** if you need to install for all users' accounts on the computer (which requires Windows Administrator privileges).
 6. Click **Next**.
 7. Select a destination folder to install Anaconda and click **Next**. Install Anaconda to a directory path that does not contain spaces or unicode characters. For more information on destination folders, see the [FAQ](#).



<https://docs.anaconda.com/free/anaconda/install/windows/>

(2) Framework Qiskit: Instalação do qiskit em um ambiente virtual via conda

1.1 Instalando o Ambiente Anaconda – Windows, macOS e Linux

Installing on macOS

⚠ Using Anaconda in a commercial setting?

Install Anaconda Distribution using either the install wizard or the command line instructions below. If you are unsure, choose the graphical install.

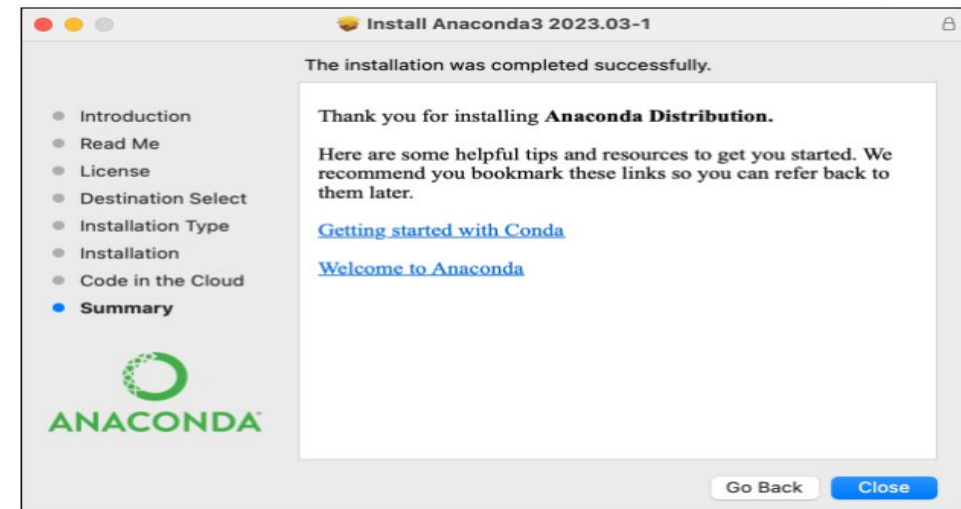
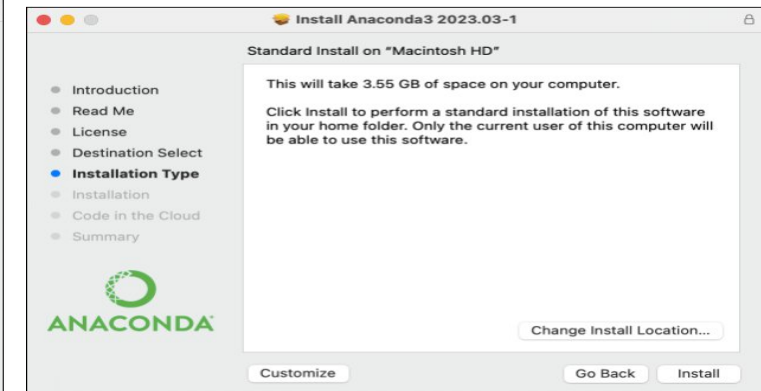
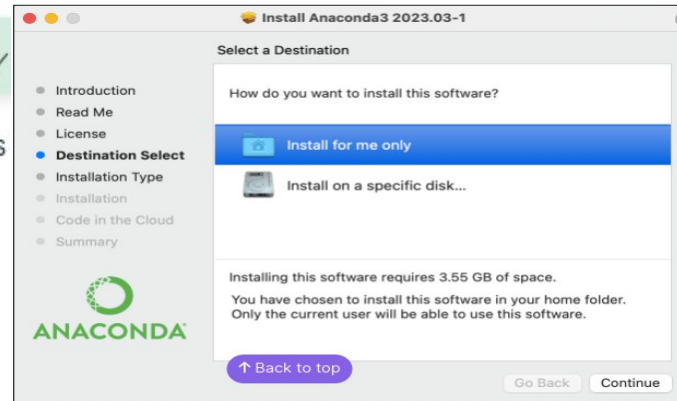
Wizard install

More of a visual learner? Watch the [Installing Anaconda \(Mac\)](#) video in the course linked below!

Anaconda install for macOS 🍏

1. Download the graphical [macOS installer](#) for your version of Python.
2. Double-click the downloaded file and click **Continue** to start the installation.
3. Answer the prompts on the Introduction, Read Me, and License screens.
4. Anaconda recommends that you choose **Install for me only**. If you do not want to install Anaconda Distribution into your home folder, select **Install on a specific disk....**

<https://docs.anaconda.com/free/anaconda/install/mac-os/>



(2) Framework Qiskit: Instalação do qiskit em um ambiente virtual via conda

1.1 Instalando o Ambiente Anaconda – Windows, macOS e Linux

Installing on Linux

⚠ Using Anaconda in a commercial setting?

Prerequisites

To use GUI (Graphical User Interface) packages with Linux, you will need to install the following extended dependencies for Qt:

Debian RedHat ArchLinux OpenSuse/SLES Gentoo

```
apt-get install libgl1-mesa-glx libegl1-mesa libxrandr2 libxrandr2 libxss1 libxcursor1 libxcomposite1 libasound2 libxi6 libxtst6
```

\$ sudo apt install libgl1-mesa-glx libegl1-mesa libxrandr2 libxrandr2 libxss1 libxcursor1 libxcomposite1 libasound2 libxi6 libxtst6

Installation

1. To download the installer, open a terminal and use the following command, depending on your Linux architecture:

Linux x86 Power8/9 AWS Graviton2/ARM64 IBMZ/LinuxOne/s390x

```
# Replace <INSTALLER_VERSION> with the version of the installer file you want to
# All installers can be found at repo.anaconda.com/archive
curl -O https://repo.anaconda.com/archive/Anaconda3-<INSTALLER_VERSION>-Linux-x86
```

3. To install, run the following command, depending on your Linux architecture:

Linux x86 Power8/9 AWS Graviton2/ARM64 IBMZ/LinuxOne/s390x

```
# Include the bash command even if you aren't using the Bash shell
# Replace ~/Downloads with the path to the installer file, if necessary
# Replace <INSTALLER_VERSION> with the version of the installer file
bash ~/Downloads/Anaconda3-<INSTALLER_VERSION>-Linux-x86_64.sh
```

4. Press Enter to review the license agreement. Then press and hold Enter to scroll.

5. Enter `yes` to agree to the license agreement.

6. Use Enter to accept the default install location, use CTRL+C to cancel the installation, or enter another file path to specify an alternate installation directory. If you accept the default install location, the installer displays `PREFIX=/home/<USER>/anaconda3` and continues the installation. It may take a few minutes to complete.

7. Anaconda recommends you enter “yes” to initialize Anaconda Distribution by running `conda init`.

8. The installer finishes and displays, “Thank you for installing Anaconda3!”

9. Close and re-open your terminal window for the installation to take effect, or enter the command `source ~/.bashrc` to refresh the terminal.

<https://docs.anaconda.com/free/anaconda/install/linux/>

(2) Framework Qiskit: Instalação do qiskit em um ambiente virtual via conda

The screenshot displays the Anaconda Navigator desktop application. The interface includes a top navigation bar with 'File' and 'Help' menus, the 'ANACONDA NAVIGATOR' logo, an 'Upgrade Now' button, and a 'Connect' button. A left sidebar contains navigation links for 'Home', 'Environments', 'Learning', and 'Community'. The main workspace is a grid of application tiles, each representing a different tool available within the Anaconda ecosystem. These tiles are organized into two rows and six columns. Each tile features a logo, the application name, a brief description, and a button to either 'Install' or 'Launch' the application. The applications shown include DataSpell, Anaconda Notebooks, VS Code, Datalore, IBM Watson Studio Cloud, Oracle Cloud Infrastructure, anaconda-toolbox, Glueviz, JupyterLab, Jupyter Notebook, Orange 3, PyCharm Professional, Qt Console, RStudio, and Spyder. The interface is clean and modern, with a light gray background and distinct color coding for each application's logo and buttons.

File Help

ANACONDA NAVIGATOR

Upgrade Now Connect

Home

Environments

Learning

Community

All applications ON base (root) Channels

DataSpell
DataSpell is an IDE for exploratory data analysis and prototyping machine learning models. It combines the interactivity of Jupyter notebooks with the intelligent Python and R coding assistance of PyCharm in one user-friendly environment.
[Install](#)

Anaconda Notebooks
Cloud-hosted notebook service from Anaconda. Launch a preconfigured environment with hundreds of packages and store project files with persistent cloud storage.
[Launch](#)

VS Code
1.85.2
Streamlined code editor with support for development operations like debugging, task running and version control.
[Launch](#)

Datalore
Kick-start your data science projects in seconds in a pre-configured environment. Enjoy coding assistance for Python, SQL, and R in Jupyter notebooks and benefit from no-code automations. Use Datalore online for free.
[Launch](#)

IBM Watson Studio Cloud
IBM Watson Studio Cloud provides you the tools to analyze and visualize data, to cleanse and shape data, to create and train machine learning models. Prepare data and build models, using open source data science tools or visual modeling.
[Launch](#)

ORACLE Cloud Infrastructure
Oracle Data Science Service
OCI Data Science offers a machine learning platform to build, train, manage, and deploy your machine learning models on the cloud with your favorite open-source tools.
[Launch](#)

anaconda-toolbox
0.4.0
[Install](#)

Glueviz
1.2.4
Multidimensional data visualization across files. Explore relationships within and among related datasets.
[Install](#)

JupyterLab
4.0.8
An extensible environment for interactive and reproducible computing, based on the Jupyter Notebook and Architecture.
[Install](#)

Jupyter Notebook
7.0.6
Web-based, interactive computing notebook environment. Edit and run human-readable docs while describing the data analysis.
[Install](#)

Orange 3
3.34.0
Component based data mining framework. Data visualization and data analysis for novice and expert. Interactive workflows with a large toolbox.
[Install](#)

PyCharm Professional
A full-fledged IDE by JetBrains for both Scientific and Web Python development. Supports HTML, JS, and SQL.
[Install](#)

Qt Console
5.5.0
PyQt GUI that supports inline figures, proper multiline editing with syntax highlighting, graphical calltips, and more.
[Install](#)

RStudio
1.1.456
A set of integrated tools designed to help you be more productive with R. Includes R essentials and notebooks.
[Install](#)

Spyder
5.4.3
Scientific Python Development Environment. Powerful Python IDE with advanced editing, interactive testing, debugging and introspection features.
[Install](#)

Anaconda Toolbox
Supercharged local notebooks. Click the Toolbox tile to install.
[Install This Toolbox](#)

Documentation

Anaconda Blog

Twitter YouTube GitHub

1.2 Instalação do qiskit em um ambiente virtual via conda

- (1) No Windows: Menu iniciar e digite: Anaconda prompt;
- (2) No macOS e Linux, apenas abra o terminal.

Crie um ambiente mínimo, apenas com o Python instalado nele.

```
conda create -n name_of_my_env python=3
```

Ative seu novo ambiente.

```
conda activate name_of_my_env
```

```
$pip install jupyterlab  
$ conda install ipykernel  
$ conda install graphviz
```

```
$ pip install qiskit
```

```
$ pip install qiskit-aer
```

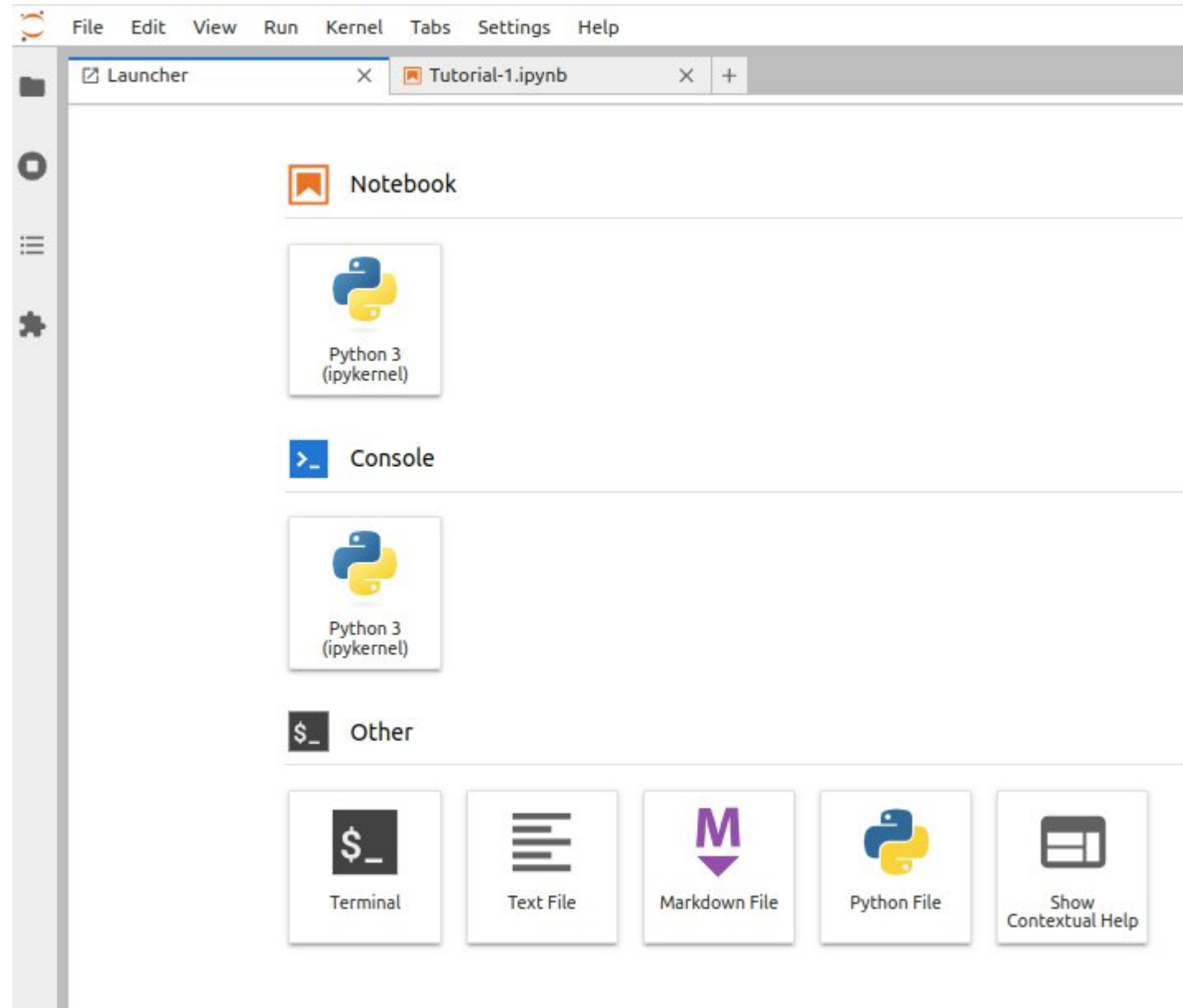
```
$ pip install qiskit[visualization]
```

```
$pip install qiskit-ibm-runtime
```

1.3 Instalação do qiskit - Teste

- (1) No Windows: Menu iniciar e digite: Anaconda prompt;
- (2) No macOS e Linux, apenas abra o terminal.

\$ jupyter-lab



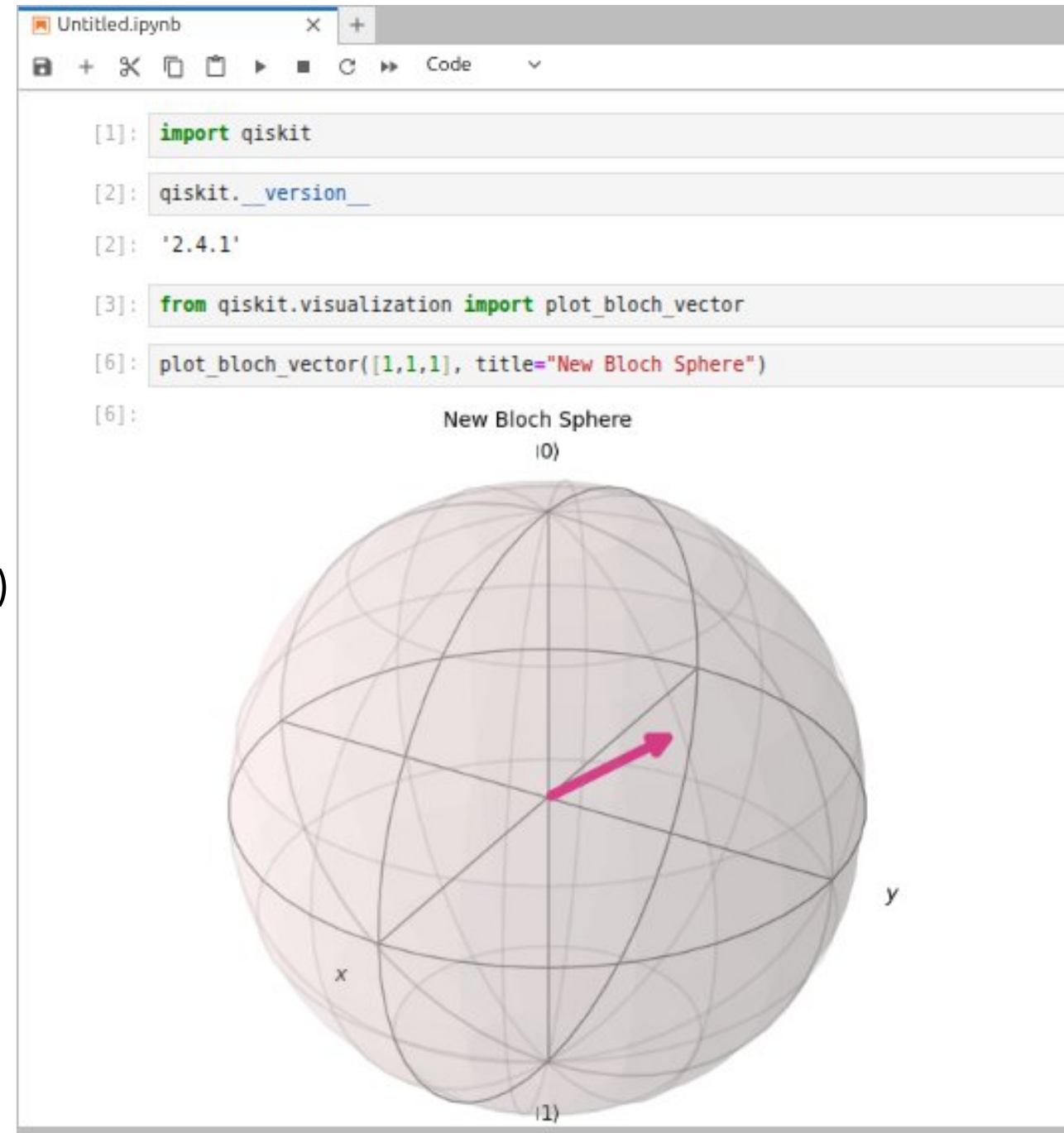
Instalação do qiskit - Teste

> import qiskit

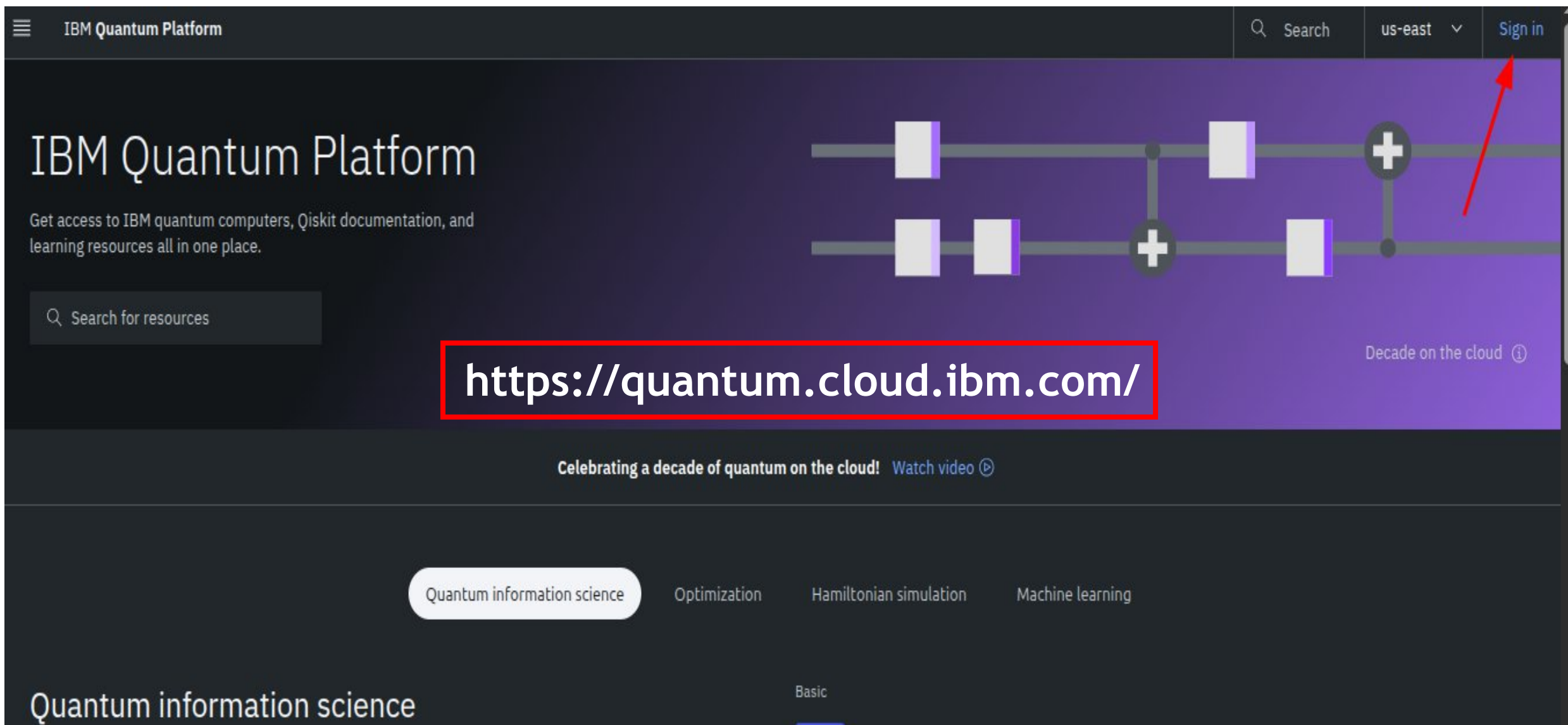
> qiskit.__version__

> from qiskit.visualization import plot_bloch_vector

> plot_bloch_vector([0,1,0], title="New Bloch Sphere")



Sign in - Plataforma IBM Quantum



The image shows the IBM Quantum Platform homepage. At the top, there is a dark navigation bar with the IBM Quantum Platform logo, a search bar, a region selector set to 'us-east', and a 'Sign in' button. A red arrow points to the 'Sign in' button. Below the navigation bar, the main content area has a dark background. On the left, the text 'IBM Quantum Platform' is displayed, followed by a sub-header: 'Get access to IBM quantum computers, Qiskit documentation, and learning resources all in one place.' Below this is a search bar with the placeholder text 'Search for resources'. In the center, there is a quantum circuit diagram with two horizontal lines representing qubits. The top line has a single-qubit gate (a square with a vertical purple bar) followed by a CNOT gate (a circle with a plus sign) controlled by the bottom line. The bottom line has a single-qubit gate followed by a CNOT gate controlled by the top line. To the right of the circuit, the text 'Decade on the cloud' is visible with an information icon. A red rectangular box highlights the URL <https://quantum.cloud.ibm.com/>. Below the main content area, there is a banner that says 'Celebrating a decade of quantum on the cloud!' with a 'Watch video' link. At the bottom, there is a horizontal menu with four items: 'Quantum information science' (which is highlighted with a white background), 'Optimization', 'Hamiltonian simulation', and 'Machine learning'. Below this menu, the text 'Quantum information science' is repeated on the left, and 'Basic' is centered above a blue bar.

IBM Quantum Platform

Get access to IBM quantum computers, Qiskit documentation, and learning resources all in one place.

Search for resources

<https://quantum.cloud.ibm.com/>

Decade on the cloud ⓘ

Celebrating a decade of quantum on the cloud! [Watch video](#)

Quantum information science Optimization Hamiltonian simulation Machine learning

Quantum information science Basic

Sign in - Plataforma IBM Quantum

IBM Quantum Platform

Search us-east

Sign in to IBM Quantum Platform

New to the platform? [Create an IBM Cloud account](#) →

Sign in with IBMid

Select a region

Washington DC (us-east) ▾

Continue →

Forgot ID?

Or

Continue with Google 

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English ▾

Sign in - Plataforma IBM Quantum

The screenshot shows the IBM Quantum Platform dashboard. A white arrow points from the top-left menu icon to the 'Criar uma Instância.' text. A red arrow points from the 'learning' instance name in the table to the explanatory text. A red box highlights the QPU cards at the bottom.

Wilson Rabelo
IBM Quantum Platform

Criar uma Instância.

Uma instância foi criada: a “learning”. Você precisa criar a sua instância com o nome da sua escolha para poder rodar projetos nos QPUs.

Celebrating a decade of quantum on the cloud! [Watch video](#)

API key [Create +](#) | [View all](#)

Instances [View all](#)

learning [CRN](#) [open](#) | 9m 47s remain

My recent workloads [View all](#)

ID	Status	Instance	Created ↓	QPU	Usage
[redacted]	✓ Completed	learning	5/6/26, 10:43 PM	ibm_marrakesh	13s
[redacted]	✓ Completed	learning	8/6/25, 11:05 AM	ibm_brisbane	1s

QPUs [View all](#)

3

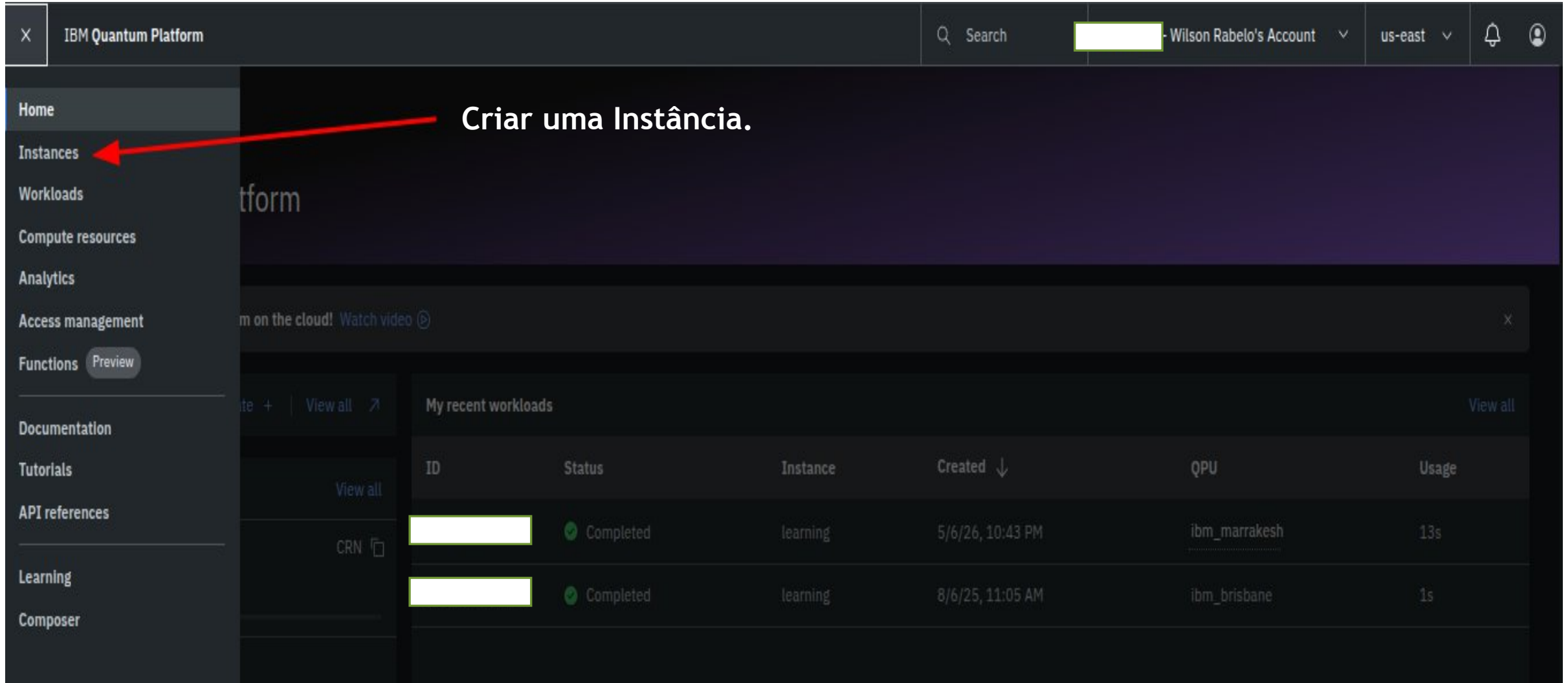
ibm_kingston
156q | us-east
Heron r2

ibm_fez
156q | us-east
Heron r2

ibm_marrakesh
156q | us-east
Heron r2

Chips, os QPU's de 156-qubits.

Sign in - Plataforma IBM Quantum



The screenshot shows the IBM Quantum Platform interface. The top navigation bar includes a search bar, a user profile dropdown (Wilson Rabelo's Account), and a region dropdown (us-east). The left sidebar contains a menu with items: Home, Instances, Workloads, Compute resources, Analytics, Access management, Functions (with a Preview button), Documentation, Tutorials, API references, Learning, and Composer. A red arrow points to the 'Instances' menu item, with the text 'Criar uma Instância.' (Create an Instance) next to it. The main content area displays a table titled 'My recent workloads' with columns: ID, Status, Instance, Created, QPU, and Usage. Two workloads are listed, both with a status of 'Completed'.

ID	Status	Instance	Created	QPU	Usage
[Redacted]	Completed	learning	5/6/26, 10:43 PM	ibm_marrakesh	13s
[Redacted]	Completed	learning	8/6/25, 11:05 AM	ibm_brisbane	1s

Sign in - Plataforma IBM Quantum

IBM Quantum Platform

Search

- Wilson Rabelo's Account

us-east

Instances

Filtered by region: us-east ⓘ

An instance is a deployment of Qiskit Runtime, needed to access quantum computers. Account owners and administrators define the instance configuration, including plan, allocated time, and QPU access.

Open (1)

13s
Total used

9m 47s
Remaining to use

10m 0s
Total available to plan

Apr 9, 2026 - May 7, 2026
Current cycle (UTC)

Search open instances by name or CRN

Create instance +

Instance	CRN	Resource group	QPUs	Cycle usage	Usage remaining	Pending workloads	Tags
learning		Default	3	13s	9m 47s	0	

Sign in - Plataforma IBM Quantum

The screenshot shows the IBM Quantum Platform dashboard for a user named Wilson Rabelo. The interface includes a top navigation bar with the platform name, a search bar, and account information. A large purple banner at the top contains the text "Precisa criar API Token." with a red arrow pointing to the "Create +" button in the "API key" section. Below this, a notification bar celebrates a decade of quantum on the cloud. The main content area is divided into two sections: "API key" and "My recent workloads". The "API key" section shows an instance named "learning" with a "CRN" button and a copy icon. A white arrow points from this "CRN" button to a text annotation at the bottom right. The "My recent workloads" section displays a table of recent jobs.

Precisa criar API Token.

API key Create + | View all ↗

Instances View all

learning CRN

open | 9m 47s remain

My recent workloads View all

ID	Status	Instance	Created ↓	QPU	Usage
	✓ Completed	learning	5/6/26, 10:43 PM	ibm_marrakesh	13s
	✓ Completed	learning	8/6/25, 11:05 AM	ibm_brisbane	1s

Essa chave, o CRN, já criada vai ser copiada para dentro da codificação Qiskit.

Hello world - Qiskit - Plataforma IBM Quantum

The screenshot displays the IBM Quantum Platform interface. At the top, the header includes the 'IBM Quantum Platform' logo, a search bar, and user account information for 'Wilson Rabelo's Account' in the 'us-east' region. Below the header, a section titled 'QPUs' shows three available quantum processors: 'ibm_kingston', 'ibm_fez', and 'ibm_marrakesh', each with a '156q | us-east' specification and a 'Heron r2' icon. A 'View all' link is present next to the 'QPUs' title. The main content area is divided into two columns. The left column, under the 'Quantum information science' tab, contains a description of Qiskit research capabilities and a list of learning resources, including a 'Basics of quantum information course' link. The right column, under the 'Basic' tab, lists several tutorials. The first tutorial, 'Hello world', is highlighted with a red arrow and includes a description: 'Tutorial • Run your first quantum circuit and view the results'. Other tutorials listed include 'Circuit library', 'Primitive inputs and outputs', and 'Combine error mitigation options with the Estimator primitive'. A 'Flameshot' watermark is visible in the bottom right corner of the interface.

IBM Quantum Platform

Search

- Wilson Rabelo's Account

us-east

QPUs [View all](#)

3

ibm_kingston 156q | us-east Heron r2

ibm_fez 156q | us-east Heron r2

ibm_marrakesh 156q | us-east Heron r2

Quantum information science Optimization Hamiltonian simulation Machine learning

Quantum information science

Conduct research with Qiskit at the level of quantum circuits and gates, leveraging tools for optimization, visualization, and execution to turn foundational ideas into experiments.

Learn

[Basics of quantum information course](#) →

Explore the research

[Disambiguating Pauli noise in quantum computers](#) ↗

[Order from chaos with adaptive circuits on quantum hardware](#) ↗

Basic

[Hello world](#)
Tutorial • Run your first quantum circuit and view the results

[Circuit library](#)
Docs • Explore ready-to-use quantum circuits and components

[Primitive inputs and outputs](#)
Docs • Learn how to define inputs and interpret results for primitives

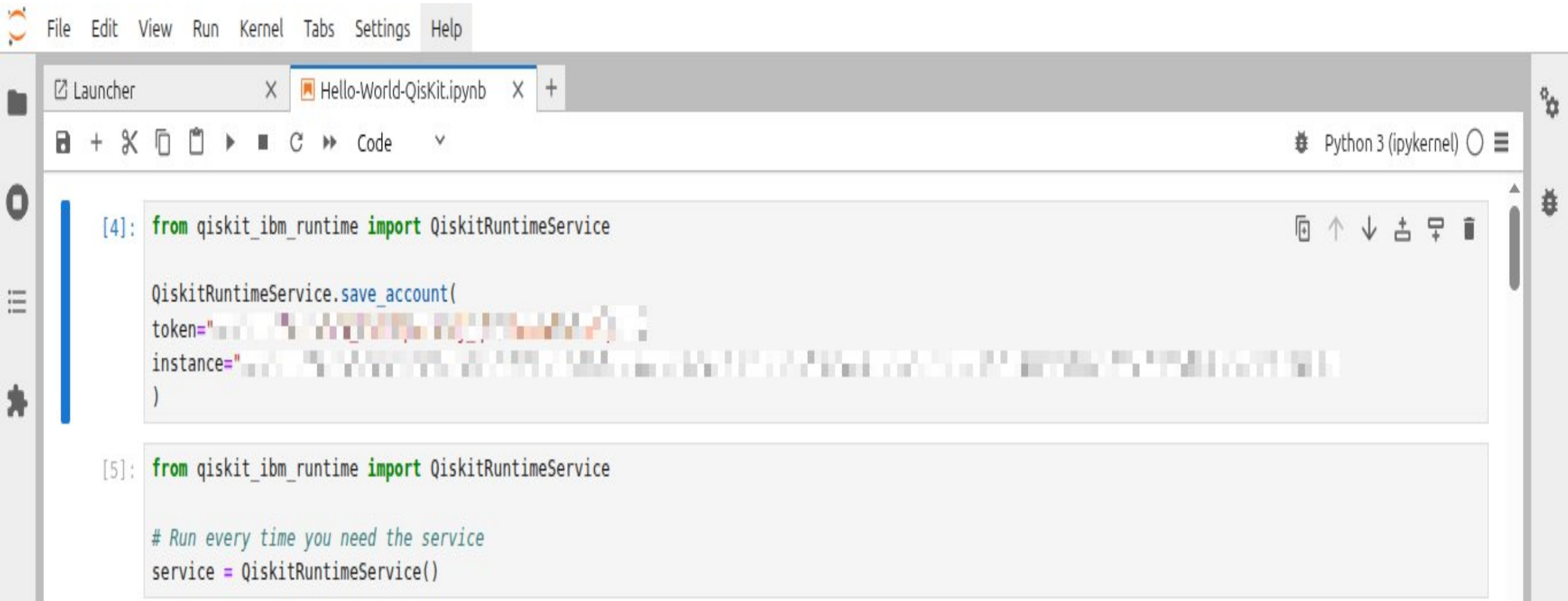
Advanced

[Combine error mitigation options with the Estimator primitive](#)
Tutorial • Improve circuit performance with error mi

Flameshot

Podem seguir esse tutorial, pois é o mesmo dos slides que virão a partir deste ponto. Apenas com alguns comentários.

Hello world - Qiskit - Plataforma IBM Quantum



The screenshot shows a Jupyter Notebook interface with a menu bar (File, Edit, View, Run, Kernel, Tabs, Settings, Help) and a toolbar. The notebook has two tabs: 'Launcher' and 'Hello-World-QisKit.ipynb'. The active cell is [4], which contains the following code:

```
[4]: from qiskit_ibm_runtime import QiskitRuntimeService

QiskitRuntimeService.save_account(
    token="REDACTED_TOKEN",
    instance="REDACTED_INSTANCE",
)
```

The next cell is [5], which contains the following code:

```
[5]: from qiskit_ibm_runtime import QiskitRuntimeService

# Run every time you need the service
service = QiskitRuntimeService()
```

Hello world - Qiskit - Plataforma IBM Quantum

File Edit View Run Kernel Tabs Settings Help

Launcher X Hello-World-QisKit.ipynb X +

Python 3 (ipykernel)

```
[6]: from qiskit import QuantumCircuit
from qiskit.quantum_info import SparsePauliOp
from qiskit.transpiler import generate_preset_pass_manager
from qiskit_ibm_runtime import QiskitRuntimeService
from qiskit_ibm_runtime import EstimatorOptions
from qiskit_ibm_runtime import EstimatorV2 as Estimator
from matplotlib import pyplot as plt
# Uncomment the next line if you want to use a simulator:
# from qiskit_ibm_runtime.fake_provider import FakeBelemV2

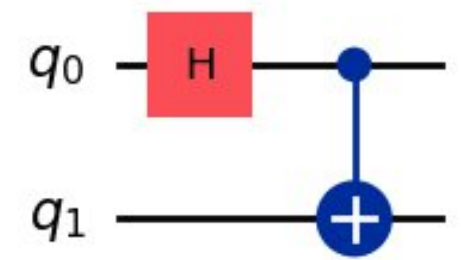
# Create a new circuit with two qubits
qc = QuantumCircuit(2)

# Add a Hadamard gate to qubit 0
qc.h(0)

# Perform a controlled-X gate on qubit 1, controlled by qubit 0
qc.cx(0, 1)

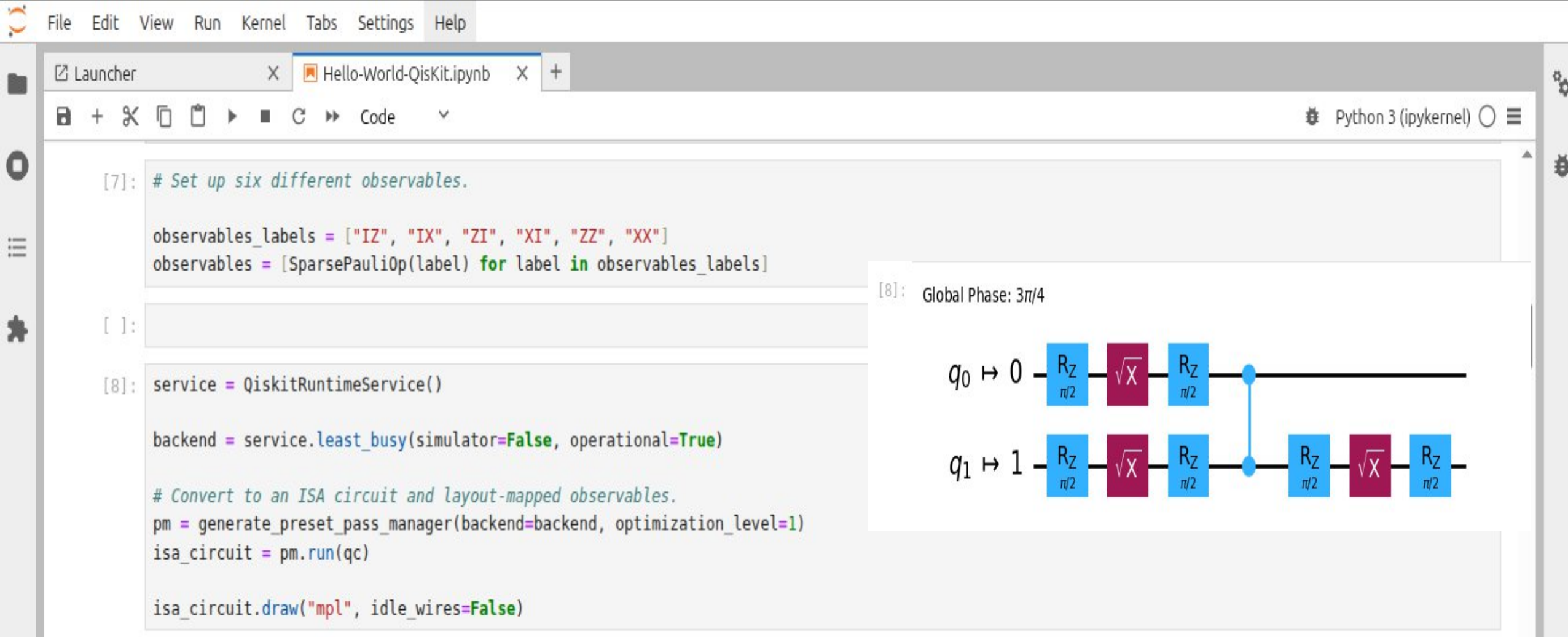
# Return a drawing of the circuit using Matplotlib ("mpl").
# These guides are written by using Jupyter notebooks, which
# display the output of the last line of each cell.
# If you're running this in a script, use `print(qc.draw())` to
# print a text drawing.
qc.draw("mpl")
```

[6]:



The diagram illustrates a quantum circuit with two qubits, q_0 and q_1 . Qubit q_0 starts with a Hadamard (H) gate, represented by a red square. This is followed by a controlled-X gate, represented by a blue dot on the q_0 line connected to a blue circle with a plus sign on the q_1 line. The q_1 line also has a controlled-X gate, represented by a blue dot on the q_1 line connected to a blue circle with a plus sign on the q_0 line.

Hello world - Qiskit - Plataforma IBM Quantum

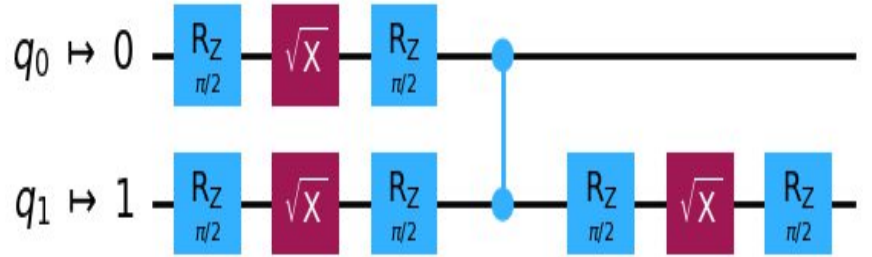


The image shows a Jupyter Notebook interface with a menu bar (File, Edit, View, Run, Kernel, Tabs, Settings, Help) and a toolbar. The notebook has two tabs: 'Launcher' and 'Hello-World-QisKit.ipynb'. The code in the notebook is as follows:

```
[7]: # Set up six different observables.  
  
observables_labels = ["IZ", "IX", "ZI", "XI", "ZZ", "XX"]  
observables = [SparsePauliOp(label) for label in observables_labels]  
  
[ ]:  
  
[8]: service = QiskitRuntimeService()  
  
backend = service.least_busy(simulator=False, operational=True)  
  
# Convert to an ISA circuit and layout-mapped observables.  
pm = generate_preset_pass_manager(backend=backend, optimization_level=1)  
isa_circuit = pm.run(qc)  
  
isa_circuit.draw("mpl", idle_wires=False)
```

Below the code, the output of the execution is displayed, showing the global phase and a quantum circuit diagram:

[8]: Global Phase: $3\pi/4$



The circuit diagram shows two horizontal lines representing qubits q_0 and q_1 . q_0 is initialized to 0 and q_1 to 1. Both qubits have a sequence of gates: $R_Z(\pi/2)$ (blue box), $\sqrt{X}$ (purple box), and $R_Z(\pi/2)$ (blue box). A CNOT gate is applied with q_0 as the control and q_1 as the target. Finally, q_1 has another sequence of gates: $R_Z(\pi/2)$ (blue box), $\sqrt{X}$ (purple box), and $R_Z(\pi/2)$ (blue box).

Hello world - Qiskit - Plataforma IBM Quantum

```
[9]: # Construct the Estimator instance.

estimator = Estimator(mode=backend)
estimator.options.resilience_level = 1
estimator.options.default_shots = 5000

mapped_observables = [
    observable.apply_layout(isa_circuit.layout) for observable in observables
]

# One job, with one circuit to run against five different observables.
job = estimator.run([(isa_circuit, mapped_observables)])

# Use the job ID to retrieve your job data later
print(f">>> Job ID: {job.job_id()}")

>>> Job ID: d7tusnaud
```


Hello world - Qiskit - Plataforma IBM Quantum

```
[ ]:
```

O problema acima foi rodado no chip físico real IBM_Marrakesh de 156-qubits, tipo de processador: Heron r2 (Heron processor).

```
[ ]:
```

```
[14]: # This is the result of the entire submission. You submitted one Pub,  
# so this contains one inner result (and some metadata of its own).  
job_result = job.result()  
  
# This is the result from our single pub, which had six observables,  
# so contains information on all six.  
pub_result = job.result()[0]
```


Hello world - Qiskit - Plataforma IBM Quantum

```
[ ]:
```

```
[15]: # Plot the result
```

```
values = pub_result.data.evs
```

```
errors = pub_result.data.stds
```

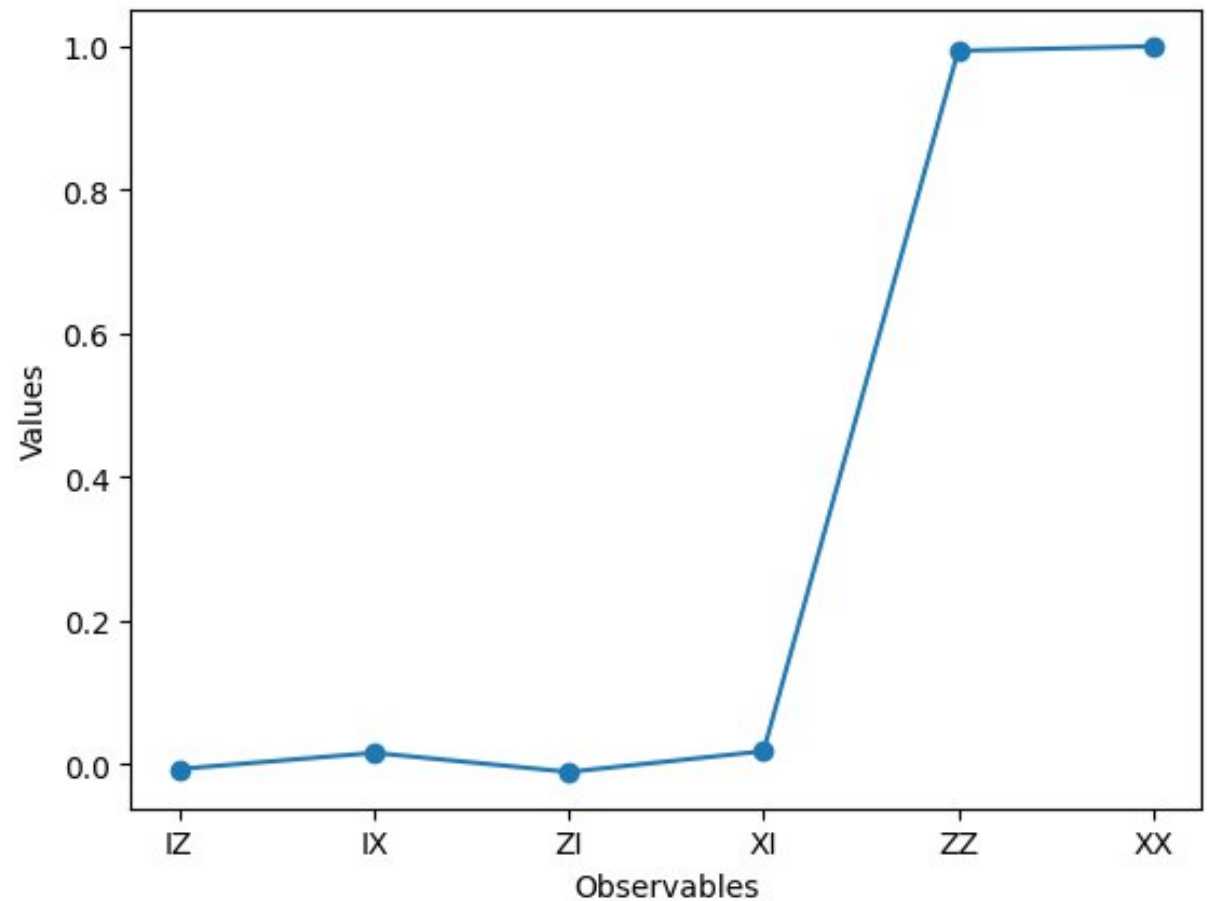
```
# plotting graph
```

```
plt.plot(observables_labels, values, "-o")
```

```
plt.xlabel("Observables")
```

```
plt.ylabel("Values")
```

```
plt.show()
```



Hello world - Qiskit - Plataforma IBM Quantum



File Edit View Run Kernel Tabs Settings Help

Launcher X Hello-World-QisKit.ipynb X +

Python 3 (ipykernel)

Alternative: run the example using a simulator - FAKE-DEVICE

FakeBelemV2 não é um chip quântico real, mas um backend simulado ("FAKE" provider) usado no Qiskit para análises e testes. Este fake-device imita o comportamento de um chip físico de 5-qubits.

Lançamento da Interface V2: O FakeBelemV2 foi introduzido como parte da transição do Qiskit para a interface BackendV2, consolidando-se amplamente em versões como o Qiskit 0.39 (lançado por volta de outubro de 2022). Hardware Original: Ele é baseado no sistema real IBM Quantum Belem, um processador de 5 qubits da família Falcon que esteve ativo e disponível na nuvem principalmente entre 2020 e 2023.

Note que, como ele é um fake-device antigo de no mínimo 3 ou 4 anos atrás observamos a qualidade dos resultados são piores do que os rodados no chip quântico real, `ibm_marrakesh`, de 156-qubits.

T1, T2 e erros de operação são as variáveis de ruído que geralmente se adota nos chips quânticos. Essas variáveis medem a qualidade e a estabilidade dos qubits. Em computação quântica, elas indicam quanto tempo a informação quântica sobrevive antes de ser destruída pelo ruído do ambiente(decoerência).

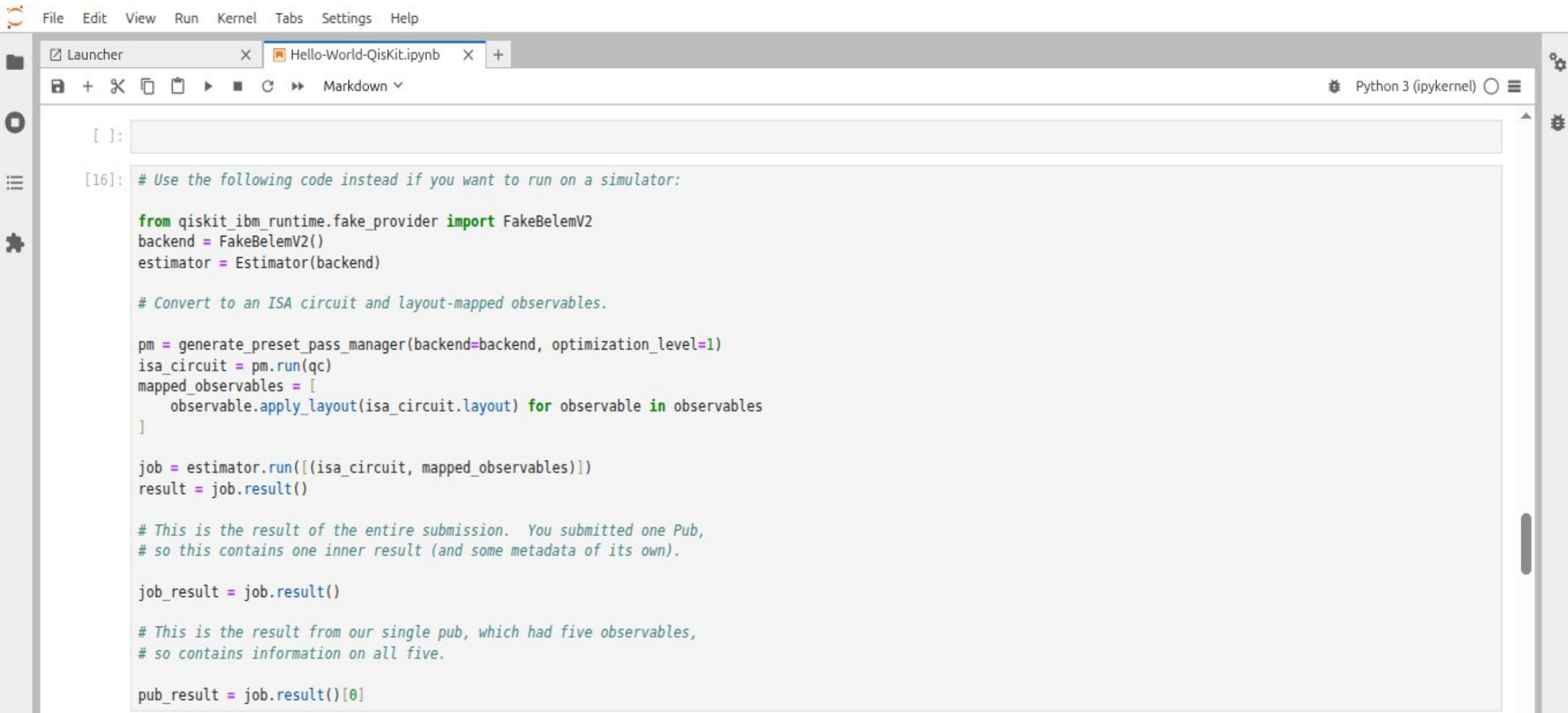
**** Tempos de Coerência (Vida Útil):**

- (1) T1 = (Tempo de Relaxação Térmica). É o tempo que um qubit leva para perder sua energia. Ou seja, o tempo de um qubit decair do estado excitado ($|1\rangle$) para o estado fundamental ($|0\rangle$).
- (2) T2 = (Tempo de Dephasing): É o tempo que o qubit mantém a sua fase quântica (a relação entre $|0\rangle$ e $|1\rangle$). Geralmente é menor que o T1.

**** Erros de Operação:**

- (1) Erro de Leitura (Readout Error): A probabilidade do sistema medir "0" quando o qubit era "1" (ou vice-versa). Ocorre devido a ruídos na eletrônica no momento de extrair o resultado.
- (2) Erro de Porta (Gate Error): A falha que ocorre ao aplicar um comando (como um "NOT" quântico). Single-qubit error: Erro em comandos aplicados a um qubit isolado. Two-qubit error (CNOT): Erro em interações entre dois qubits (geralmente muito mais alto que o de um qubit).

Hello world - Qiskit - Plataforma IBM Quantum



The image shows a Jupyter Notebook interface with a menu bar (File, Edit, View, Run, Kernel, Tabs, Settings, Help) and a toolbar. The active tab is 'Hello-World-QisKit.ipynb'. The code cell contains the following Python code:

```
[ ]:
[16]: # Use the following code instead if you want to run on a simulator:

from qiskit_ibm_runtime.fake_provider import FakeBelemV2
backend = FakeBelemV2()
estimator = Estimator(backend)

# Convert to an ISA circuit and layout-mapped observables.

pm = generate_preset_pass_manager(backend=backend, optimization_level=1)
isa_circuit = pm.run(qc)
mapped_observables = [
    observable.apply_layout(isa_circuit.layout) for observable in observables
]

job = estimator.run([(isa_circuit, mapped_observables)])
result = job.result()

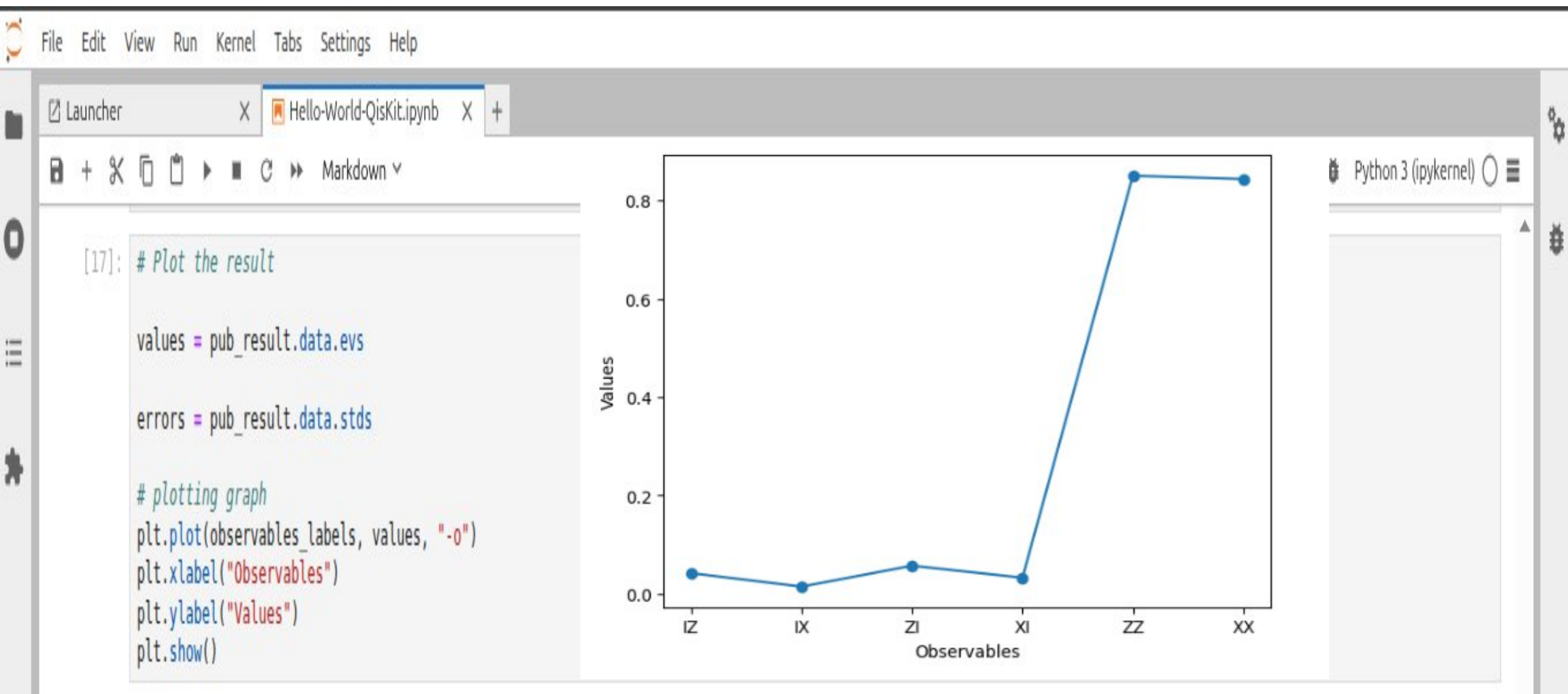
# This is the result of the entire submission. You submitted one Pub,
# so this contains one inner result (and some metadata of its own).

job_result = job.result()

# This is the result from our single pub, which had five observables,
# so contains information on all five.

pub_result = job.result()[0]
```

Hello world - Qiskit - Plataforma IBM Quantum



Referências

Para este capítulo foi retirado das referências abaixo: figuras, fotos e exercícios e demais códigos com algumas alterações.

[1] M. A. Nielsen and I. L. Chuang, Quantum Computation and Quantum Information, Cambridge University Press, (2000).

[2] Qiskit. Disponível em: <https://qiskit.org/>

[3] IBM Quantum. Disponível em:
<https://quantum.cloud.ibm.com/>

[4] Laboratório de Computação Quântica-ITEC-UFPA (LCQ).
Disponível em: <https://sites.google.com/ufpa.br/quantumlab/lcq>